A Dual-Backbone Hybrid Deep Learning Framework with Squeeze-and-Excitation Attention and Grad-CAM Explainability for Multi-Class Brain Tumor MRI Classification

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***Abstract*—Accurate brain tumor classification from magnetic resonance imaging (MRI) is essential for treatment planning, yet manual interpretation remains time-consuming and subject to inter-observer variability. This paper proposes a dual-backbone convolutional neural network that fuses EfficientNetB3 and MobileNetV2 features through a Squeeze-and-Excitation (SE) channel-attention block for classifying glioma, meningioma, pituitary tumor, and no-tumor MRI scans. The preprocessing pipeline combines Gaussian–bilateral denoising with contrast-limited adaptive histogram equalization (CLAHE) to suppress noise while preserving diagnostically relevant contrast. The model was trained progressively through frozen-backbone warm-up, partial fine-tuning, and full end-to-end optimization with cosine learning-rate scheduling on 7,153 MRI images. The proposed network achieved 99.72% validation accuracy and 99.16% test accuracy, with macro precision, recall, F1-score, and ROC-AUC values of 0.9914, 0.9917, 0.9915, and 0.9997, respectively. Ablation analysis showed that direct backbone fusion without SE attention reached only 95.35% accuracy, whereas the complete attention-enhanced model achieved 99.16%, confirming the importance of adaptive channel recalibration. Grad-CAM was used to visualize class-relevant tumor regions and improve prediction transparency. These findings demonstrate that the proposed attention-guided dual-backbone framework provides accurate, interpretable, and computationally practical brain tumor classification for clinical decision-support applications.**

***Index Terms*—brain tumor classification, magnetic resonance imaging, hybrid deep learning, EfficientNet, MobileNetV2, squeeze-and-excitation attention, explainable AI, Grad-CAM**

I. Introduction

Brain tumors are among the most serious neurological diseases, and accurate identification of tumor subtypes, including glioma, meningioma, and pituitary tumor, is essential for prognosis and treatment planning [1]. Magnetic resonance imaging (MRI) is widely used for brain tumor assessment because of its superior soft-tissue contrast. Nevertheless, manual interpretation is time-consuming, requires specialist expertise, and remains vulnerable to inter- and intra-observer variability, particularly in subtle or ambiguous cases [2]. These challenges have encouraged the development of automated deep-learning systems for MRI-based tumor classification. Convolutional neural networks (CNNs), trained from scratch or adapted from ImageNet-pretrained architectures such as VGG, ResNet, and DenseNet, have achieved strong performance on public brain MRI datasets [3]. Recent research has increasingly emphasized both computational efficiency and model interpretability. EfficientNet and MobileNet architectures provide compact feature extractors with favorable accuracy–complexity trade-offs, while Squeeze-and-Excitation (SE) blocks improve feature discrimination through channel-wise recalibration [4], [5]. In parallel, explainable AI methods such as Grad-CAM [6], Grad-CAM++, and LIME help identify image regions that contribute to predictions, which is important for improving clinical confidence in automated systems [7], [8]. Hybrid models that fuse complementary features from multiple backbones have shown potential to improve classification performance without relying on extremely deep or computationally expensive architectures. In particular, combinations of EfficientNet- and MobileNet-family networks with SE-based attention can integrate detailed and lightweight representations while supporting efficient inference [9], [10]. However, many existing studies provide limited analysis of preprocessing, component contributions, inference cost, or visual interpretability.

To address these limitations, this paper proposes a dual-backbone framework for four-class brain tumor MRI classification. Its main contributions are as follows:

- An EfficientNetB3–MobileNetV2 hybrid architecture is introduced, where complementary feature embeddings

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are fused and recalibrated using an SE channel-attention block.
- A preprocessing pipeline combines Gaussian–bilateral denoising with contrast-limited adaptive histogram equalization (CLAHE) to suppress noise and enhance diagnostically relevant contrast.
- A three-stage training strategy is employed, consisting of frozen-backbone warm-up, partial fine-tuning, and full end-to-end optimization with cosine learning-rate scheduling.
- A component-wise ablation study compares both individual backbones, direct feature fusion, and SE-enhanced fusion to quantify the contribution of each component.
- The model is evaluated using class-wise precision, recall, F1-score, confusion matrices, ROC-AUC, model complexity, inference latency, and Grad-CAM visualizations across all four classes.

II. Related Work

Automated brain tumor classification from MRI has been studied extensively using both classical machine learning and deep learning approaches. Early classical pipelines relied on handcrafted features, such as texture descriptors, combined with classifiers such as support vector machines before CNN-based transfer learning became the dominant paradigm [3]. Transfer-learning studies that fine-tune pretrained backbones such as VGG-16, ResNet50, and EfficientNetB0 on brain tumor MRI datasets have consistently reported high accuracy on the glioma, meningioma, pituitary, and no-tumor four-class classification task, establishing this formulation as a common benchmark [10], [11].

A growing number of studies specifically target lightweight and hybrid architectures. Dash and Sisodia incorporated Squeeze-and-Excitation blocks into a lightweight CNN, SEIMB-Net, improving channel-wise feature recalibration while maintaining a low parameter count [11]. Closely related to the design adopted in this paper, AbdelAziz *et al.* proposed SECNN-MNet, which combines SE-based convolutional feature extraction with MobileNet depthwise separable convolutions and reported strong performance on four-class brain tumor classification [5]. Similarly, Celik *et al.* combined MobileNetV3 and EfficientNetB7 features through a co-attention ensemble mechanism, showing that feature fusion can outperform individual backbones [12]. These findings motivate the dual-backbone EfficientNetB3–MobileNetV2 fusion adopted in this work, where a more expressive EfficientNet branch and a lightweight MobileNet branch are integrated through a shared SE-attention gate.

Explainability is another important research direction in medical image classification. Since deep models are often opaque, recent studies combine high-accuracy classifiers with visualization techniques such as Grad-CAM, Grad-CAM++, LIME, and saliency maps to support clinical interpretability [7], [8]. Rasool *et al.* proposed CNN-TumorNet, using LIME to explain model predictions for tumor diagnosis from MRI images [8]. Rastogi *et al.* introduced XAI-BT-EdgeNet, a dual-branch edge-aware SE network that integrates multiple XAI techniques, including LIME, Grad-CAM, Grad-CAM++, and saliency maps, to improve clinical trust [13]. Other recent hybrid CNN approaches also use Grad-CAM to visualize salient tumor regions while maintaining high classification accuracy [14]. Consistent with this direction, the present work integrates Grad-CAM into the evaluation pipeline and additionally reports model size and inference latency, which are less consistently documented in prior studies.

Preprocessing also plays an important role in MRI-based tumor classification. Noise reduction and contrast enhancement can improve feature visibility before CNN-based feature extraction [2]. Therefore, this study combines Gaussian filtering, bilateral filtering, and contrast limited adaptive histogram equalization (CLAHE) as part of the preprocessing pipeline. Unlike studies that treat preprocessing as a fixed step, the proposed framework evaluates the complete pipeline end-to-end with classification, ablation, efficiency, and explainability analysis.

III. Materials and Proposed Methodology

A. Dataset

The dataset [15] contains 7,153 T1-weighted brain MRI images from four classes: glioma, meningioma, no tumor, and pituitary tumor. It includes 1,621 glioma, 1,775 meningioma, 2,000 no-tumor, and 1,757 pituitary images, representing 22.7%, 24.8%, 28.0%, and 24.6% of the dataset, respectively. The maximum-to-minimum class ratio is 1.23, indicating a nearly balanced distribution; nevertheless, class weighting was applied during training.

Using a fixed random seed, the dataset was divided at the file level into training, validation, and test subsets. The training set contains 5,005 images (70%), while the validation and test sets contain 1,073 and 1,075 images (15% each), respectively. Class proportions were preserved across all splits.

B. Preprocessing Pipeline

Each image was converted from BGR to RGB and processed using two-stage denoising. A 3×3 Gaussian blur was followed by a bilateral filter with $d = 9$ and $\sigma_{color} = \sigma_{space} = 75$ to reduce acquisition noise while preserving tumor boundaries. Contrast enhancement was then performed using CLAHE with a clip limit of 2.0 and an 8×8 tile grid. CLAHE was applied to the luminance channel in LAB color space.

The processed images were resized to 224×224 pixels and normalized to the $[0, 1]$ range. During training, online augmentation included rotation up to 15°, width and height shifts of 10%, horizontal flipping, and zooming up to 10%. No stochastic augmentation was applied to the validation or test sets. Fig. 1 illustrates the complete preprocessing sequence.

C. Proposed Hybrid Architecture

The proposed architecture, shown in Fig. 2, processes the same $224 \times 224 \times 3$ input through two parallel ImageNet-pretrained backbones.

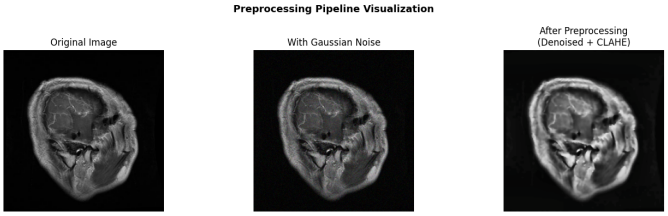


Fig. 1. Preprocessing pipeline showing the original MRI slice, noisy input, and the output after Gaussian–bilateral denoising and CLAHE enhancement.

1) EfficientNetB3 Branch: The EfficientNetB3 classification head was removed, and the extracted semantic features were processed using global average pooling, a 256-unit dense layer with ReLU activation, batch normalization, and dropout of 0.3.

2) MobileNetV2 Branch: The MobileNetV2 branch follows the same projection sequence, producing a complementary 256-dimensional lightweight feature representation with batch normalization and dropout of 0.3.

The two embeddings are concatenated into a 512-dimensional joint representation. An SE attention block then performs channel-wise recalibration by compressing the feature vector to 64 units with ReLU activation and expanding it back to 512 units with sigmoid activation. The resulting weights are multiplied element-wise with the fused representation, allowing the network to emphasize informative channels. The attended features are passed through a 256-unit dense layer with ReLU, batch normalization, and dropout of 0.4, followed by a 128-unit dense layer with ReLU and dropout of 0.3. A four-unit softmax layer produces probabilities for glioma, meningioma, no tumor, and pituitary tumor:

$$\hat{y} = \mathrm{Softmax}(Wz + b), \tag{1}$$

where z denotes the attended feature vector and $\hat{y}$ is the predicted class-probability distribution. The complete model contains approximately 13.99 million parameters and requires about 161 MB of storage.

IV. Training Protocol

The model was trained on an NVIDIA T4 GPU using Adam, categorical cross-entropy, a batch size of 32, and a three-stage progressive schedule.

A. Frozen and Partial Fine-Tuning

In Trial 1, both pretrained backbones were frozen, and only the fusion, SE attention, and classification layers were trained for 10 epochs with a learning rate of 1×10^{-4}. This stage achieved 85.09% validation accuracy. In Trial 2, the final 30 layers of each backbone were unfrozen and fine-tuned for another 10 epochs using 1×10^{-5}. Validation accuracy remained 85.09%, indicating limited improvement from partial fine-tuning.

TABLE I
Comparison with baseline classification models.

Model	Acc.	Prec.	Rec.	F1	Params
Simple CNN	82.70	0.8480	0.8180	0.8154	0.46M
ResNet50	72.28	0.7315	0.7156	0.7101	24.11M
EfficientNetB3 [16]	60.93	0.6451	0.6065	0.6119	11.18M
MobileNetV2 [17]	92.00	0.9215	0.9177	0.9176	2.59M
DenseNet121	90.88	0.9116	0.9077	0.9065	7.30M
VGG16	90.14	0.8995	0.8991	0.8981	14.85M
InceptionV3	92.28	0.9203	0.9214	0.9204	22.33M
Xception	**94.70**	**0.9461**	**0.9458**	**0.9457**	21.39M
InceptionResNetV2 [18]	91.81	0.9159	0.9173	0.9160	54.73M
Proposed	**97.86**	**0.9783**	**0.9781**	**0.9780**	**14.00M**

B. Full End-to-End Training

In Trial 3, all layers were trained for up to 30 epochs using a base learning rate of 3×10^{-4}. A three-epoch linear warm-up was followed by cosine learning-rate decay. Gradient clipping with `clipnorm`=1.0 and balanced class weights were applied. Early stopping monitored validation accuracy, while ReduceLROnPlateau monitored validation loss with a factor of 0.5 and patience of 3 epochs. The best checkpoint reached 99.72% validation accuracy at epoch 26 and was selected for final testing.

V. Results and Discussion

A. Baseline Model Comparison

The proposed architecture was compared with a scratch-trained CNN and several ImageNet-pretrained baselines, including ResNet50, EfficientNetB3, MobileNetV2, DenseNet121, VGG16, InceptionV3, Xception, and InceptionResNetV2. As shown in Table I, Xception was the strongest baseline with 94.70% accuracy and an F1-score of 0.9457. The proposed EfficientNetB3–MobileNetV2 model with SE attention achieved 97.86% accuracy, with macro precision, recall, and F1-score of 0.9783, 0.9781, and 0.9780, respectively. This 3.16 percentage-point gain over Xception demonstrates the benefit of complementary dual-backbone features and SE-based channel recalibration.

B. Overall Performance

Table II summarizes the final results. The proposed model achieved 99.16% test accuracy, with macro precision, recall, F1-score, and ROC–AUC of 0.9914, 0.9917, 0.9915, and 0.9997, respectively, indicating strong and balanced class discrimination. Figure 3 shows that only 9 of 1,075 test images were misclassified, with glioma achieving 1.00 recall and only minor confusion among the remaining classes. As shown in Figure 4, all classes achieved at least 0.98 precision, recall, and F1-score. The ROC curves in Figure 5 further confirm excellent separability across the four classes.

C. Comparison Across Training Trials

Table III compares the three training trials. Freezing both backbones in Trial 1 achieved 85.09% validation accuracy, indicating that ImageNet-pretrained features alone were insufficient to fully separate MRI-specific tumor textures. Trial

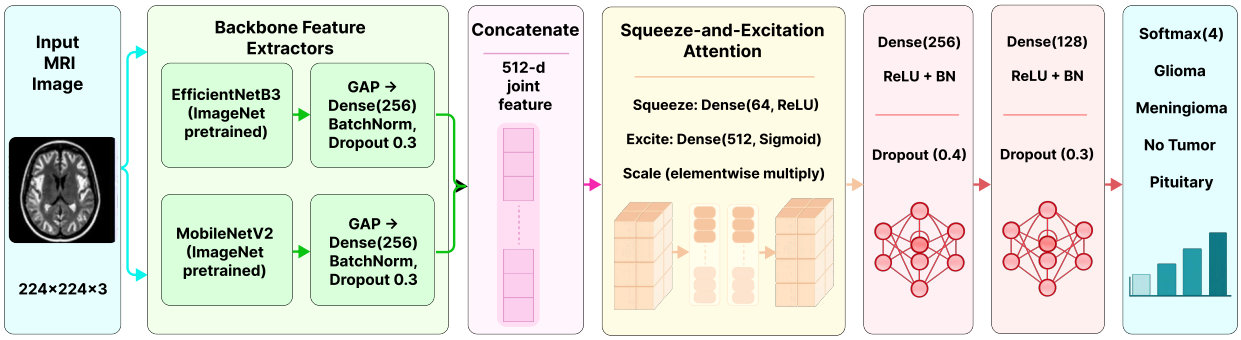


Fig. 2. Proposed dual-backbone hybrid architecture. EfficientNetB3 and MobileNetV2 branches extract complementary feature embeddings, which are fused through concatenation, recalibrated by a Squeeze-and-Excitation attention block, and classified using a dense classification head.

TABLE II
FINAL MODEL PERFORMANCE

Validation Accuracy	Validation Loss	Test Accuracy	Test Loss	Precision (Macro)	Recall (Macro)	F1-Score (Macro)	Macro ROC–AUC
99.72%	0.0139	99.16%	0.0493	0.9914	0.9917	0.9915	0.9997

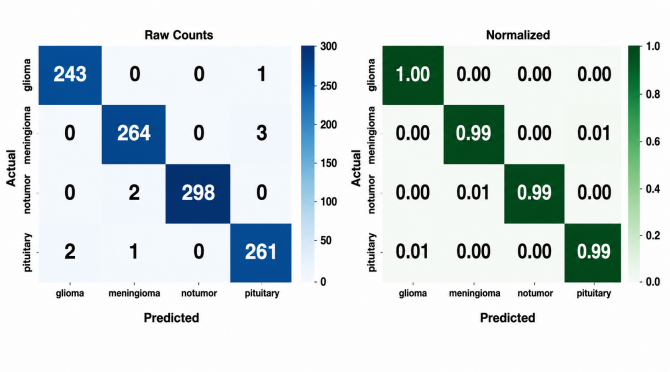


Fig. 3. Confusion matrix on the held-out test set: (left) raw counts and (right) row-normalized values.

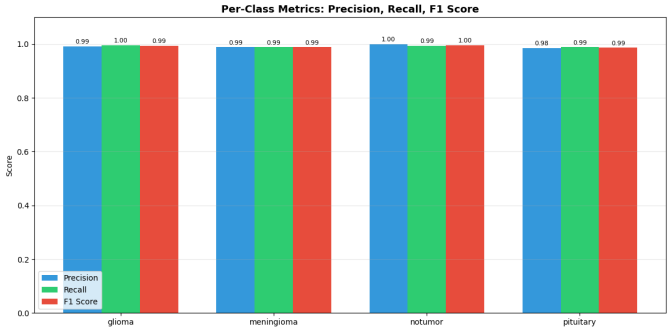


Fig. 4. Per-class precision, recall, and F1-score on the test set.

2, where only the top 30 layers of each backbone were fine-tuned using a lower learning rate, produced the same validation accuracy of 85.09%, suggesting limited benefit from shallow fine-tuning under a conservative learning rate. In contrast, Trial 3 achieved 99.72% validation accuracy using full end-to-end training with a cosine-annealed learning-rate schedule.

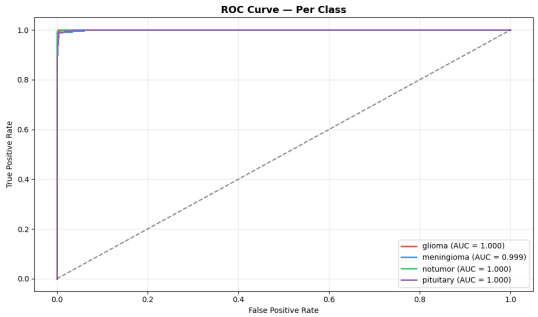


Fig. 5. Per-class ROC curves on the test set. The macro-average AUC is 0.9997.

TABLE III
COMPARISON OF THE THREE TRAINING TRIALS

Trial	Strategy	Epochs	Val. Acc.
Trial 1	Frozen backbones	10	85.09%
Trial 2	Partial fine-tune	10	85.09%
Trial 3	Full fine-tune, cosine LR	up to 30	**99.72%**

This improvement of approximately 14.6 percentage points confirms that adapting the entire network to the MRI domain is essential for achieving strong classification performance.

D. Ablation Study: Component Contribution

To quantify the contribution of each architectural component, three variants were evaluated using the same preprocessing, class weighting, and optimizer settings as the full model. Each ablation model was trained for 15 epochs with cosine-annealed learning. The evaluated variants were EfficientNetB3 alone, MobileNetV2 alone, and dual-backbone fusion without SE attention. Table IV summarizes the test results.

TABLE IV
ABLATION STUDY OF THE PROPOSED ARCHITECTURE

Variant	Acc.	Prec.	Rec.	F1
EfficientNetB3 only	89.49%	0.9065	0.8946	0.8869
MobileNetV2 only	97.30%	0.9725	0.9737	0.9730
Fusion without SE	95.35%	0.9526	0.9532	0.9529
Full model (proposed)	**99.16%**	**0.9914**	**0.9917**	**0.9915**

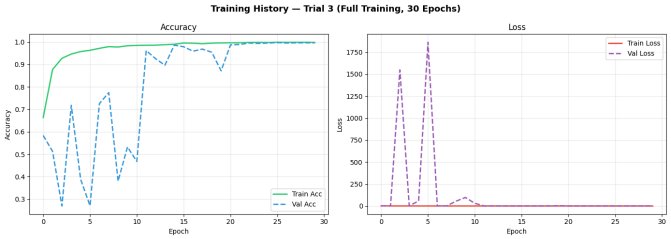


Fig. 6. Training and validation performance of the proposed model, showing rapid convergence within approximately 15 epochs.

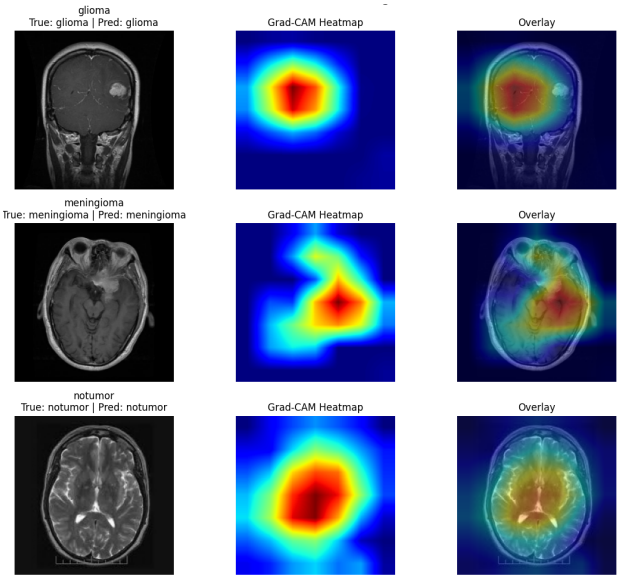


Fig. 7. Grad-CAM visualizations for representative test images from the four classes. Each row shows the original MRI scan, the Grad-CAM heatmap, and the overlay. Rows from top to bottom represent glioma, meningioma, no-tumor, and pituitary tumor.

MobileNetV2 outperformed EfficientNetB3 as a standalone backbone, achieving 97.30% versus 89.49% accuracy. Direct fusion without attention reached 95.35%, whereas SE-based recalibration improved accuracy to 99.16%, yielding gains of 1.86 and 3.81 percentage points over MobileNetV2 and naive fusion, respectively. This confirms the importance of channel-wise attention in the proposed fusion strategy.

Although the ablation variants were trained for 15 epochs, Fig. 6 shows that the proposed model had largely converged within this period, making the comparison informative despite the unmatched training budgets.

E. Explainability: Grad-CAM Analysis

Grad-CAM was applied to the final convolutional feature layer, `top_conv`, to examine whether predictions relied on clinically plausible image regions. Fig. 7 presents representative examples from all four classes, including the original MRI, activation map, and overlay. The visualizations emphasize tumor-associated regions in glioma, meningioma, and pituitary cases, while the no-tumor example exhibits weaker and more diffuse activation. These findings provide qualitative evidence that the model primarily attends to anatomically relevant structures rather than background artifacts.

Fig. 8 shows representative correctly classified test images from the four classes together with their predicted labels and confidence scores. All four samples were correctly classified with confidence scores of at least 0.98, further demonstrating the discriminative capability of the proposed model.

F. Model Efficiency and Inference Latency

In addition to classification accuracy, practical deployment requires low inference latency and manageable model size. The final model contains 13,996,851 total parameters, of which 13,873,900 are trainable, and the saved model occupies approximately 161 MB. Averaged over 50 runs after a warm-up pass, single-image inference on an NVIDIA T4 GPU required 107.52 ± 27.37 ms, with a minimum latency of 79.05 ms. Batch inference over 32 images required 664.43 ms in total, corresponding to approximately 20.76 ms per image. These results indicate that the proposed model is suitable for near-real-time per-slice screening. However, since the timing was measured on a shared GPU environment, inference speed should be re-evaluated on dedicated clinical hardware before deployment.

G. Comparison with Related Work

Table V compares the proposed model with representative hybrid and attention-based methods for four-class brain tumor MRI classification. The EfficientNetB3–MobileNetV2–SE framework achieved 99.16% test accuracy, demonstrating performance competitive with recent approaches. Unlike many related studies, the proposed pipeline jointly incorporates denoising and contrast enhancement, progressive training, Grad-CAM-based interpretation, and measured inference latency.

H. Limitations

The study has several limitations. The dataset was obtained from a single public source without external multi-center validation, so robustness to clinical domain shift remains uncertain. Because patient identifiers were unavailable, file-level splitting may allow related slices from the same patient to appear across subsets and potentially inflate performance. Evaluation was also based on a single fixed split, while the ablation models used a shorter training budget than the final model. In addition, the framework analyzes 2-D slices only, and Grad-CAM explanations were not quantitatively validated against expert annotations. Future work will focus on patient-disjoint multi-center evaluation, repeated validation, 3-D mod-

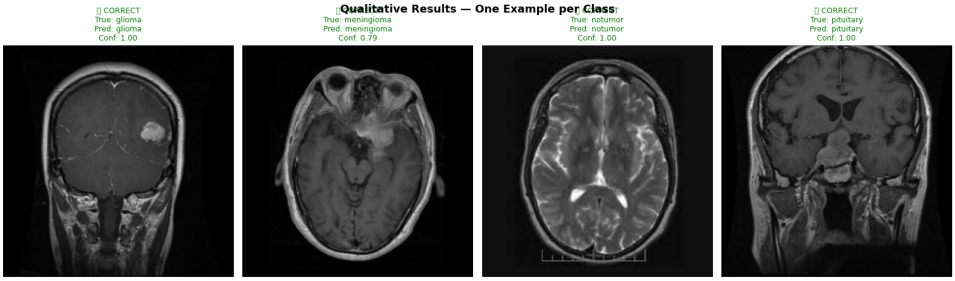


Fig. 8. Representative correctly classified test images, one from each class, with the predicted label and confidence score.

TABLE V
COMPARISON WITH REPRESENTATIVE HYBRID AND ATTENTION-BASED APPROACHES

Approach	Backbone(s)	Acc.
SECNN-MNet [5]	SE-CNN + MobileNet	99.6%
Ensemble attention [12]	MobileNetV3 + EfficientNetB7	98.94%*
Squeezed-Eff-Net [4]	SqueezeNet + EfficientNetB0	99.08%
CNN + XGBoost + Grad-CAM [14]	Custom CNN	99.77%
Proposed	**EfficientNetB3 + MobileNetV2 + SE**	**99.16%**

*Evaluated on the Figshare dataset rather than the Nickparvar four-class dataset used in this work; included only for architectural comparison.

eling, matched ablation settings, and quantitative explainability assessment.

VI. CONCLUSION

This study introduced a dual-backbone framework that combines EfficientNetB3 and MobileNetV2 through Squeeze-and-Excitation attention for four-class brain tumor MRI classification. The proposed model achieved 99.16% test accuracy, a macro F1-score of 0.9915, and a macro-average ROC-AUC of 0.9997. Ablation results showed that simple feature concatenation was insufficient, while SE-based channel recalibration substantially improved classification performance. Grad-CAM visualizations further indicated that the model focused on clinically relevant tumor regions, supporting its potential as a decision-support system. Future work will investigate external multi-center validation, patient-level cross-validation, 3-D volumetric analysis, quantitative assessment of explanation quality, and model compression for resource-constrained clinical deployment.

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